

Loss Functions

The executable objective is grouped into Part-structure information, Visual evidence, candidate correction, route interpretation, and auxiliary semantic refinement. The coefficients below are the values in [configs/progressive_candidate_correction.yaml1](#).

Common definitions

For probability p , target t , and valid mask v , Dice loss is computed per sample and channel:

$$\mathcal{L}_{\text{Dice}}(p, t; v) = 1 - \text{mean}_{b,c} \frac{2 \sum_x v(x)p(x)t(x) + \epsilon}{\sum_x v(x)(p(x) + t(x)) + \epsilon}$$

Class-response and semantic classification use generalized cross-entropy with $q = 0.7$:

$$\mathcal{L}_{\text{GCE}}(p_y) = \frac{1 - p_y^q}{q}$$

All masked losses are divided by valid-mask mass. Semantic and Part Map class weights are calculated from training-mask frequencies:

$$\tilde{w}_k = (f_k + 10^{-8})^{-1/2}, \quad w_k = \frac{\tilde{w}_k}{K^{-1} \sum_j \tilde{w}_j}$$

The normalized background weight is multiplied by 0.25.

Part-structure information

Let B be Fish Region and P the five-class Part Map:

$$\mathcal{L}_B = 0.80 (\mathcal{L}_{\text{BCE}}(B) + 0.50 \mathcal{L}_{\text{Dice}}(B)),$$

$$\mathcal{L}_P = 0.80 (\mathcal{L}_{\text{CE}}(P) + 0.35 \mathcal{L}_{\text{Dice}}(P))$$

Fish Region uses the full image. Part Map uses reviewed fish pixels. The Head-to-Tail Direction and Zone Map are continuous maps derived from the predicted geometry and Part Map and do not add separate weighted terms.

Visual evidence

For Redness, Shape, and Lesion:

$$\ell_R = \mathcal{L}_{\text{BCE}}(E_R) + 0.25 \mathcal{L}_{\text{Dice}}(E_R),$$

$$\ell_S = \mathcal{L}_{\text{BCE}}(E_S) + 0.25 \mathcal{L}_{\text{Dice}}(E_S),$$

$$\ell_L = \mathcal{L}_{\text{BCE}}(E_L) + 0.35 \mathcal{L}_{\text{Dice}}(E_L).$$

The positive Visual evidence component is

$$\mathcal{L}_{E+} = 0.25 \ell_R + 0.45 \ell_S + 0.55 \ell_L + 0.12 \mathcal{L}_{\text{BCE}}(E_{\cup}, M),$$

where $E_{\cup} = \max(E_R, E_S, E_L)$. The Unaffected Surface component is

$$\mathcal{L}_{EU} = 0.55 (\mathcal{L}_{\text{BCE}}(E_U) + 0.35 \mathcal{L}_{\text{Dice}}(E_U))$$

The class-dependent soft targets and valid masks are listed in [SUPERVISION.pdf](#).

Candidate correction

Candidate classification uses

$$\mathcal{L}_{\text{cls}} = 0.15 \mathcal{L}_{\text{GCE}}(R_0) + 0.30 \mathcal{L}_{\text{GCE}}(N) + 0.25 \mathcal{L}_{\text{GCE}}(C_g) + 0.90 \mathcal{L}_{\text{GCE}}(C),$$

where C_g is the intermediate gated response. All four terms use annotated symptom pixels.

The accompanying candidate terms are

$$\mathcal{L}_{\text{proto}} = 0.40 \mathcal{L}_{\text{GCE}}(P_{\text{sim}}) + 0.12 \text{mean}(1 - \cos(e, p_y)),$$

$$\mathcal{L}_{\text{part}} = 0.45 \mathcal{L}_{\text{BCE}}(H_{\text{part}}),$$

where P_{sim} is symptom-prototype similarity, e is the normalized visual embedding, p_y is the target-class prototype, and H_{part} is the part-conditioned abnormality response.

Let p_{fg} be the symptom-foreground response and p_{seed} the maximum class-seed response:

$$\mathcal{L}_{\text{fg}} = 0.35 (\mathcal{L}_{\text{BCE}}(p_{\text{fg}}, M) + 0.25 \mathcal{L}_{\text{Dice}}(p_{\text{fg}}, M)) + 0.25 \mathcal{L}_{\text{BCE}}(p_{\text{seed}}, 1)$$

Suppression and coverage terms are

$$\begin{aligned} \mathcal{L}_{\text{sup}} = & 0.20 \text{ mean}_{T_V}(P_{\text{class}}) + 0.12 \text{ mean}_{T_V}(P_{\text{seed}}) + 0.20 \text{ mean}_{T_V}(P_{\text{body}}) \\ & + 0.10[\rho_{\text{body}} - 0.12]_+^2 + 0.08[\rho_{\text{fg}} - 0.18]_+^2 + 0.16[\rho_{\text{caudal}} - 0.62]_+^2. \end{aligned}$$

The three ρ terms are Fish-Region body-lesion coverage, Fish-Region foreground coverage, and caudal-region caudal-necrosis coverage.

Route interpretation

Let O be Route assignment, Q Spatial Support, L_r the route-specific class response, and Z Routed semantic logits. The route loss is

$$\begin{aligned} \mathcal{L}_{\text{route}} = & 0.75 \mathcal{L}_{\text{GCE}}(Z) + 0.35 \mathcal{L}_{\text{CE}}(O, T_O) + 0.25 \mathcal{L}_{\text{CE}}(O, P) \\ & + 0.55 (\mathcal{L}_{\text{BCE}}(Q, T_Q) + 0.25 \mathcal{L}_{\text{Dice}}(Q, T_Q)) \\ & + 0.65 \mathcal{L}_{\text{local}} + 0.025 \mathcal{L}_{\text{preserve}} + 0.010 \mathcal{L}_{\Delta} \\ & + 0.030 \mathcal{L}_{\text{consistent}} + 0.020 \mathcal{L}_{\text{close}} + 0.040 \mathcal{L}_{\text{unaffected}}. \end{aligned}$$

The local route term is

$$\mathcal{L}_{\text{local}} = 0.25 \mathcal{L}_{\text{body}} + 0.30 \mathcal{L}_{\text{fin}} + 0.30 \mathcal{L}_{\text{caudal}} + 0.15 \mathcal{L}_{\text{mouth}}$$

The none class receives an Unaffected Surface validity weight of 0.10. $\mathcal{L}_{\text{preserve}}$ keeps routed probabilities close to C where route confidence is low, using confidence softness 0.70. $\mathcal{L}_{\Delta}$ is the confidence-weighted absolute route-logit change. $\mathcal{L}_{\text{consistent}}$ penalizes Spatial Support outside its assigned part, and $\mathcal{L}_{\text{close}}$ penalizes holes relative to a 5 x 5 soft closing.

Semantic output and final refinement

For semantic logits V , define

$$\ell_{\text{sem}}(V) = \mathcal{L}_{\text{GCE}}(V) + 0.35 \mathcal{L}_{\text{Dice,fg}}(V)$$

With refined logits V_r and native logits V_n :

$$\mathcal{L}_{\text{sem}} = 0.45 (\ell_{\text{sem}}(V_r) + 0.35 \ell_{\text{sem}}(V_n))$$

Final refinement uses

$$\begin{aligned} \mathcal{L}_{\text{refine}} = & 0.40 (0.35 \mathcal{L}_{\text{SmoothL1,SDF}} + 0.25 \mathcal{L}_{\text{Dice,fg}} \\ & + 0.05 \mathcal{L}_{\text{curvature}} + 0.025 \mathcal{L}_{\text{TV}} + 0.025 \mathcal{L}_{\text{bilateral}}). \end{aligned}$$

The predicted and target signed distances are transformed by $\tanh(\cdot/4)$ and evaluated in the five-pixel boundary band. Curvature and total variation use an edge-aware weight formed from Shape, Lesion, and image edges. The bilateral term preserves strong image edges.

Stage-dependent objective

The route loss shares the Candidate/Route multiplier shown below. The eleven columns reproduce the executable stage schedule.

Epochs	Fish	Part	E_+	E_U	Part-cond.	Candidate/Route	Prototype	Semantic	Foreground	Refinement	Suppression
1-25	1.00	1.00	0	0	0	0	0	0	0	0	0
26-50	0.05	0.05	1.00	1.00	0.35	0	0	0	0.10	0	0.05
51-60	0	0	0	0.15	0.35	0.25	1.00	0.15	0.20	0	0.10
61-85	0	0	0	0.10	0.90	1.00	0.85	0.45	0.55	0.10	0.40
86-100	0	0	0	0.10	0.45	0.35	0.35	0.50	0.35	1.00	0.30
101-120	0	0	0	0.15	0.80	0.85	0.60	0.65	0.55	0.45	0.45

The implementation is in `losses/objective.py`.